MATHEMATICS

University *of* NORTH ALABAMA

# Spring 2025 Math Problem of the Week: Problem #8

Due noon Monday, March 10
Open to all UNA undergraduates

---

*Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, March 10 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!*

---

**Problem #8:** Let $A, B$ be $2 \times 2$ matrices with real entries satisfying $(AB - BA)^n = \mathcal{I}_2$ for some positive integer $n$. Prove that $n$ is even and $(AB - BA)^4 = \mathcal{I}_2$.